

iv) Find the quadratic equation whose one root is $-2 + \sqrt{3}$ [1]

v) Find the value of $3^{1/3}, 3^{1/9}, 3^{1/27}, \dots$ [1]

19. a) Evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{e^{\cos x} - 1}{\frac{\pi}{2} - x}$ [2]

b) If the quadratic equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root then prove that the root must be either $\frac{pq' - p'q}{q - q'}$ or $\frac{q - q'}{p' - p}$ [3]

GROUP-B [3×8=24]

20. a) A square is drawn by joining the mid points of the sides of a given square. A third square is drawn inside the second square in the same way and this process continues indefinitely. If a side of the first square is 16 cm, determine the sum of the areas of all the square. [3]

b) The sum of an infinite number of terms of a GP is 23 and sum of their square is 69. Determine the first term and the common ratio. [2]

c) If the roots of the equation $ax^2 + cx + c = 0$ are in the ratio $m:n$, then prove that: $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + \sqrt{\frac{c}{a}} = 0$ [3]

21. a) Define symmetric difference of two sets A & B. For any sets A and B, prove that $A \Delta B = (A \cup B) - (A \cap B)$ [1+3]

b) Prove that for any two statements p and q

i) $\sim(p \vee q) \equiv \sim p \wedge \sim q$ [2]

ii) $\sim(p \vee (\sim q)) \equiv (\sim p) \wedge q$ [2]

22. Evaluate the following limits:

a) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x}$ [2]

b) $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x+5} - \sqrt{x})$ [2]

c) $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta - \sin \theta}{\theta - \frac{\pi}{4}}$ [2]

d) If $f(x) = \frac{ax+b}{x-1}$, $\lim_{x \rightarrow 0} f(x) = 2$ and $\lim_{x \rightarrow \infty} f(x) = 5$, find the value of $f(4)$ [2]

Best of luck!

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